Venkata Lakshmi Pasupuleti

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PROFESSIONAL SUMMARY

Software engineer and AI agent builder with 8+ years of software development and production code across multi-agent orchestration systems, cloud migration, and enterprise-scale data pipelines. Built **AgentProbe** - a 3-agent LangGraph runtime with CI-integrated evaluation passes and multi-provider LLM routing - and **Leasel** - an 8-agent LangGraph platform with retrieval-augmented generation and human-in-the-loop escalation. At Entrata (PropTech, 100K+ MAU), led infrastructure design, application design, and cloud migration on AWS across 120+ Agile sprints. Improving agent accuracy through prompt engineering and RAG pipeline optimization. Delivered technical workshops to 600+ students on AI applications and software systems. AWS Bedrock and AgentCore: actively evaluating. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Multi-Agent Orchestration | AI Agent Development | Retrieval-Augmented Generation | Prompt Engineering | Production Code Delivery | LLM Evaluation & Eval Passes | AWS Bedrock / AgentCore | Workshop & Partner Enablement

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an **agentic mission-planning system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) - delivering a risk-aware compliance strategy: compliant missions export, non-compliant route to rejection with grounded reasons.
- Implemented a **RAG layer** (retrieval-augmented generation over FAA Part 107 corpus) - every compliance check functions as a hypothesis test cited to source regulation; designed a backend-agnostic retriever with ChromaDB upgrade path.
- Engineered **validation guardrails** enforcing FAA altitude limits, battery budgets, and geometric no-fly-zone checks - blocking non-compliant missions before export; improved agent accuracy through iterative prompt engineering and retrieval tuning.
- Parsed natural-language requests into structured flight plans using quantitative photogrammetry math; delivered agentic system to production across marketing, real estate, and commercial survey operations.

George Mason University - College of Engineering

Aug 2025 - May 2026

Graduate Teaching Assistant (Part-time)

- Delivered technical **workshops** to **600+ students** across multiple MIS 303 sections - Python programming, database design, and AI application fundamentals with weekly office hours and lab-based instruction.
- Developed supplementary exercises, code walkthroughs, and rubric-aligned feedback; coordinated workshop content delivery with lead instructor across all sections.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer (Remote)

- **Shipped production code across 20+ full-stack modules** (PHP, Python, React, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ MAU - 99.5% uptime over 6+ consecutive months.
- **Shipped first production AI pipeline** on 1.6M real financial records (Python, PyTorch, OpenAI API) - automated data processing replacing manual Excel workflows; output validation gates and quality controls at launch.
- **Led microservices infrastructure design** using Docker - 50% improvement in system scalability, 30% reduction in deployment time; converted 20% of product modules into service-based architecture.
- **Led cloud migration** of resident data to AWS S3 with Libsodium-encrypted pipelines - 100% PHI compliance, zero security incidents; led application design decisions across 20+ complex production schemas.
- Optimized 200+ SQL queries (30% performance gain); eliminated 80% of manual workload through Python/PL-SQL automation; mentored 5+ junior engineers across 300+ peer code reviews.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built full-stack property management modules (PHP, Python, PostgreSQL); processed 1.6M HUD/Tax Credit records; cloud migration of client documents to AWS S3; redesigned document storage database architecture.

PROJECTS

AgentProbe: Agentic AI Testing Toolkit

Live on GitHub

Deployed a 15-endpoint FastAPI backend running 100% locally by building a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool-use patterns, inter-agent state handoffs, and multi-provider routing (Anthropic, OpenAI, Groq, Ollama). Built a CI-integrated **5-metric LLM-as-judge evaluation passes** pipeline (factual accuracy, hallucination, retrieval, completeness, compliance) - every commit scored automatically, regressions blocked before merge. React dashboard delivers trace-level observability across concurrent agent test suites: per-run metrics, regression drift, and retrieval quality. MCP-compatible tool interface design.

Python · LangChain · LangGraph · FastAPI · ChromaDB · Anthropic API · OpenAI API · Groq · Ollama · React · Docker · GitHub Actions

Leasel: AI Document Intelligence & Support Investigation Platform

2026

Built a three-surface enterprise AI knowledge platform: (1) **Document Intelligence** - OCR-fallback ingestion, retrieval-augmented generation, 100% citation correctness, human-review gate on regulated decisions; (2) **Support Investigation** - 8-agent LangGraph pipeline with conditional routing and HITL escalation producing ranked, evidence-backed root-cause hypotheses; (3) Tenant Self-Service with hard eligibility boundaries. Core design rule enforced in code: LLM extracts and reasons; deterministic logic or human makes every final call. Achieved 93.6% precision on 1,240 lease PDFs across 5 US states.

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · Streamlit

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University — 2017

SKILLS

Languages: Python, PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash

AI & Agents: LangChain, LangGraph, AI Agents, Multi-Agent Orchestration, Tool Calling, ChromaDB, RAG Pipelines (Retrieval-Augmented Generation), LLM Evaluation, Prompt Engineering, HITL, Human-in-the-Loop, Output Validation, LLM-as-Judge, Eval Frameworks, Production AI, MCP, AWS Bedrock (evaluating), AWS AgentCore (evaluating), FastAPI, OpenAI API, Anthropic API, Groq, Ollama

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Application Design, Infrastructure Design, Cloud Migration, Software Development, Microservices, Distributed Systems, PostgreSQL, MySQL, Oracle, RabbitMQ, Git, Linux, Agile/Scrum

ML & Data: PyTorch, Scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, Statistical Modeling, NumPy, Pandas, spaCy, BERT, Tableau, Power BI